

arc42 Documentation Links

An aerial photograph of a cityscape during sunset or sunrise. The sky is a mix of blue and orange. In the foreground, there are several modern, multi-story buildings with white facades and many windows. A prominent bridge or walkway connects two of these buildings. A rainbow is visible on the left side of the image, arching over a street. In the background, a city skyline is visible, including a tall chimney emitting smoke. The overall atmosphere is hazy and warm.

Inter-Chapter Relationships in arc42

Note: arc42 is a template for software architecture documentation with 12 chapters. Understanding how these chapters connect is crucial for creating coherent, consistent documentation.

Note: These links create traceability, showing how architectural decisions flow from requirements to implementation.

Note: Each connection tells a story about your system's architecture and design rationale.

Why Links Matter in arc42

Creating Architectural Traceability:

- Combines decision with problem with solution
- Ensures consistency across documentation
- Enables impact analysis when changes occur
- Provides justification for architectural choices

Benefit: Stakeholders can follow the architectural reasoning from high-level goals to technical details.

Link 1: Solution Strategy → Quality Scenarios

Connection:

- **From:** Chapter 5 - Solution Strategy
- **To:** Chapter 11 - Quality Scenarios

Purpose:

- Shows how strategic decisions address specific quality requirements
- Provides concrete examples of how strategy is implemented
- Demonstrates the practical application of architectural principles

Example:

Creating a plugin architecture for payment providers supports **QS_MT_1**, where **QS_MT_1** is the id of a quality scenario specified in chapter 11: **Quality Scenario _ Maintenance _ 1**.

Link 2: Quality Tree Leaves → Quality Scenarios

Connection:

- **From:** Quality Tree (within Chapter 9 - Quality Requirements)
- **To:** Chapter 11 - Quality Scenarios

Purpose:

- Transforms abstract quality attributes into verifiable scenarios
- Supports common understanding of qualities between stakeholder and architect

Example:

Reliability is more detailed specified by the quality scenario **QS_R_1**. The quality scenario could be: "Under normal load (< 500 sessions) a node failure of the database is detected within 5 seconds."

Link 3: Quality Tree Leaves → Cross-Cutting Concepts

Connection:

- **From:** Quality Tree (within Chapter 9 - Quality Requirements)
- **To:** Chapter 10 - Cross-Cutting Concepts

Purpose:

- Shows how concepts are set to support quality goals
- Ensures consistency in achieving quality goals

Example:

The maintainability is supported by using an efficient logging library as a cross cutting concept (**CCC_1**). This ensures that timestamps and other environment parameter are included into the log message. After all this will help us investigating bugs.

Link 4: Solution Strategy → Main Requirements

Connection:

- **From:** Chapter 5 - Solution Strategy
- **To:** Chapter 4 - Requirements Overview (Functional Requirements)

Purpose:

- Justifies why certain architectural approaches were chosen
- Shows alignment between technical strategy and business needs
- Documents trade-offs made between competing requirements

HINT: Use ids for the main requirements as well (e.g.: REQ_1) when you create this relations

Link 5: Solution Strategy → Quality Goals

Connection:

- **From:** Chapter 5 - Solution Strategy
- **To:** Chapter 9 - Quality Requirements (Quality Goals)

Purpose:

- Demonstrates how strategic decisions prioritize quality attributes
- Shows architectural focus areas based on business priorities
- Documents the rationale for quality trade-offs

Note: When you document solution strategy in table format there is already a column named goal/requirement in place, that can be use to establish the link.

Link 6: Glossary

In the glossary you can define abbreviations or acronyms. In the text you can make use of them as a link up to glossary.

Link 7: Arbitrary

In case you need to apply other links than those described above, feel free to do so.